Remote Desktop Services (RDS)

Windows Server Roles and Components for Remote Access

What is Remote Desktop Services?

RDS is a Windows Server role that enables users to access applications and desktops remotely over a network connection, providing centralized computing resources.

Client Device → RD Gateway → RD Connection Broker

RD Web Access ↕ RD Session Host

RD Gateway

Purpose: Provides secure internet access to internal RDS resources

Functions:

- Secure HTTPS/SSL connections
- Firewall traversal
- Authentication and authorization
- Connection policies
- Remote access from internet

RD Connection Broker

Purpose: Manages and directs user connections to appropriate session hosts

Functions:

- Load balancing connections
- Session reconnection
- Personal desktop assignments
- High availability coordination
- User session management

RD Session Host

Purpose: Hosts applications and desktop sessions for remote users

Functions:

- Run applications remotely
- Multi-user sessions
- Resource management

RD Web Access

Purpose: Provides web-based portal for accessing remote resources

Functions:

- Web-based application portal
- Single sign-on experience
- Application publishing

- Published applications
- Full desktop sessions

- Browser-based access
- User customization

Common Deployment Scenarios

Single Server

- All roles on one server
- Small business (5-25 users)
- Simple setup
- Lower cost
- Limited scalability

Multiple Servers

- Roles distributed across servers
- Medium business (25-100 users)
- Better performance
- Improved reliability
- Easier maintenance

High Availability

- Redundant components
- Enterprise (100+ users)
- Load balancing
- Failover clustering
- Maximum uptime

Licensing Requirements

Windows Server Licensing

- Windows Server Standard/Datacenter
- Per-core licensing model
- Minimum 16 cores per server
- Virtual machine rights included
- RDS role included in OS

RDS Client Access Licenses (CALs)

- Required for each user or device
- Per User CAL vs Per Device CAL
- External connector alternative
- Must match Windows Server version
- Separate purchase required

Implementation Best Practices

Security Considerations

- Use SSL certificates
- Implement Network Level Authentication
- Configure connection policies
- Enable audit logging
- Use strong authentication
- Regular security updates

Performance Optimization

- Size servers appropriately
- Monitor resource utilization
- Configure session limits
- Optimize network bandwidth
- Use connection broker for load balancing
- Regular performance monitoring

Benefits of RDS

- **Centralized Management:** Applications and data in one location
- **Cost Effective:** Extend hardware life, reduce client requirements

Common Use Cases

- **Remote Workers:** Access company applications from home
- **Branch Offices:** Centralized applications for multiple locations

- **Security:** Data remains on server, not on client devices
- **Accessibility:** Access from anywhere with internet
- **Scalability:** Add users without new hardware
- **Maintenance:** Update applications once, affects all users

- **BYOD Environments:** Personal devices accessing corporate apps
- **Legacy Applications:** Modernize access to older software
- **Temporary Staff:** Quick access without full workstation setup
- **High Security:** Keep sensitive data in data center

Key Takeaway: RDS enables secure, centralized remote access to applications and desktops. Each role serves a specific purpose: Gateway for internet access, Broker for connection management, Session Host for running applications, and Web Access for user portals.